

Dong Payton Pei

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SUMMARY

Software engineer building local-first AI systems, developer tools, and production mobile applications. Former Nokia and Ford co-op developer/tester and current master's student in Systems Science and Engineering at the University of Ottawa, with hands-on experience in Rust/Tauri, TypeScript/React, Python, SwiftUI, Kotlin/Jetpack Compose, and LLM tool calling.

EDUCATION

University of Ottawa

Master of Systems Science and Engineering

Hangzhou / Ottawa

2024 — Present

Carleton University

Bachelor of Computer Science

Hangzhou / Ottawa

2017 — 2023

PROJECTS

pipi-shrimp-agent

github.com/mammut001/pipi-shrimp-agent

- Architected a local-first desktop AI agent using Rust, Tauri 2, React, TypeScript, and SQLite, with multi-provider streaming, structured tool calling, project-scoped memory, and local code and file execution.
- Implemented visual workflow orchestration, CDP browser automation, multi-agent collaboration, context compression, and native Typst PDF/SVG rendering through a Rust-backed Tauri command layer.

Conveyor

github.com/mammut001/Conveyor

- Built a self-hosted Python control plane connecting Telegram and Feishu to Codex CLI on a VPS, executing development tasks inside isolated Git worktrees with diff, apply, discard, and cancellation controls.
- Designed a persistent SQLite job queue and safety layer with operator allowlists, explicit confirmation gates, output redaction, audit logging, diagnostics, reminders, and optional developer-tool integrations.

Resume Generator

resume-tailor.paytonpei.top

- Built and deployed a full-stack React, TypeScript, and Node.js resume workspace supporting PDF/text import, job-description tailoring, structured editing, and Typst-backed PDF/SVG export.
- Added Zod-validated backend routes, privacy-conscious observability, Docker deployment, and automated testing with Vitest and Playwright.

Focus Mint

apps.apple.com/us/app/focus-mint-focus-timer-study/id6759029810

- Shipped a SwiftUI productivity app on the App Store for iPhone, iPad, and Apple Watch, featuring focus sessions, earnings tracking, analytics, custom modes, and multilingual localization.
- Implemented StoreKit monetization, WidgetKit and Live Activity experiences, and bidirectional iPhone-Apple Watch synchronization using WatchConnectivity.

WORK EXPERIENCE

Software Tester Co-op

Aug 2020 — Aug 2021

Nokia Canada

Hangzhou / Ottawa

- Enhanced JavaScript/XML code to improve code coverage and robustness.
- Produced Confluence design documentation for system architecture and data flows.

Software Developer Co-op

Apr 2020 — Aug 2020

Ford Motor Canada

Hangzhou / Ottawa

- Participated in Agile development, wrote C++11 unit tests with GoogleTest, and used Jenkins in the CI/CD workflow.

RESEARCH

Low-Light RAW Image Restoration with Bridge U-Net and DDBM

2026

- Collaborated on an extreme low-light RAW image restoration study using paired SID and ELD images, comparing Bridge U-Net, fine-tuning, few-shot meta-learning, and a DDBM-inspired feature-space approach.
- Investigated and documented a memory-efficient DDBM-Feature variant using time-conditioned decoding, selective bridge interpolation over deep encoder features, and three-step deterministic inference; evaluated it at 26.12 dB PSNR and 0.807 MS-SSIM on 598 SID Sony test images.

SKILLS

- **Languages:** Python, TypeScript, JavaScript, Rust, Swift, Kotlin, C++, Java, Bash
- **Frameworks:** React, Node.js, Tauri 2, SwiftUI, Jetpack Compose, Firebase, Tailwind CSS
- **AI & Systems:** LLM APIs, Tool Calling, Embeddings, SQLite, CDP Browser Automation, WatchConnectivity, StoreKit 2
- **Testing & Delivery:** PyTest, Vitest, Playwright, GoogleTest, Jenkins, Docker
- **Additional:** MCP, Function Calling, Agent Harness, Skills, SSE, IPC, Sandbox, Multi-Agent